

Hosting a Static Website on S3

1. Static vs. Dynamic Websites:

- A **static website** contains fixed content that doesn't change unless you update the files.
- A **dynamic website** interacts with users, allowing data to be updated or fetched dynamically.

2. Why Choose S3 for Static Websites?

- S3 is an ideal choice for hosting static websites because it's cost-effective and easy to manage.
- Since dynamic content is not supported by S3 natively, S3 is best suited for static websites where the content doesn't change frequently.

3. Steps to Host a Static Website on S3:

- Open your **S3 bucket** in the AWS Management Console.
- Go to the **Properties** tab.
- Under the **Static website hosting** section, configure the settings based on your requirements (such as index document and error document).
- Save the changes. AWS will generate a **website endpoint (DNS name)** for you.

4. Testing Your Website:

- You can test the website using any browser by accessing the **DNS name** provided by S3.
- To create the website content, you can use simple **HTML code**. Save the website files on your computer, such as `index.html` or other necessary files.
- Upload the files to your S3 bucket.

5. Setting Permissions:

- By default, the files in the S3 bucket will not be publicly accessible. You need to make them publicly accessible by changing permissions.
- Open the **bucket** and navigate to the **Permissions** tab.
- Create a **bucket policy** in **JSON format** to allow access to your objects. A bucket policy typically includes three important elements:
 1. **Effect**: The permission (allow or deny).
 2. **Action**: The actions that can be performed on the objects (like `s3:GetObject`).
 3. **Resource**: The ARN (Amazon Resource Name) specifying the objects or bucket.
- If you're unsure how to write a bucket policy, you can use tools like the **S3 Bucket Policy Generator** by searching for it on Google.
- Based on your requirements, select the appropriate options and generate the policy. Copy the JSON policy and paste it into the **Bucket Policy Editor** in the S3 Permissions tab.

6. Accessing Your Static Website:

- Once the permissions are set, your static website will be live and accessible via the S3 **website endpoint (DNS)**.

- You can then visit your static website hosted on S3 by entering the **website URL** in your browser.